

# Attested Transition Execution for DAZE-Style Simulator Backdoors

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USENIX Security 2026 Poster Abstract

## Abstract

Reinforcement-learning (RL) agents are increasingly trained on third-party simulators, making the simulator a supply-chain trust anchor. DAZE shows that a malicious simulator wrapper can implant reward-free backdoors by changing what transition the learner observes after it submits an action without changing the action itself, so action-level logging cannot detect it [1]. Prior RL backdoor work mainly studies policy/agent triggers, competitive backdoors, or sparse targeted attacks [2, 3, 4]; we instead protect the simulator data plane. We study the complementary defense problem under an explicit trust boundary: the simulator core, configuration/assets, random-number state, observation and termination code, reward inputs, and actuator transform are measured and trusted, while the wrapper is untrusted. *Attested Transition Execution* (ATE) accepts a learner-visible update only when an approved measured step relation, exact replay, or an enrolled receipt service vouches for the full transition record. The headline result is zero admitted tampered updates: on original DAZE Hopper-v5 and Reacher-v5 PPO, attack ASR reaches at least 0.8435 and 0.9843, while ATE restores clean-level ASR and matched benign-attested return with ASR-lift 0 and utility 1.0000 over 3.94M defended receipt checks; on SAC Reacher-v5, ASR drops from 0.9996 to 0 with zero forged updates.

## Claim and Contributions

**Claim.** For an approved measured simulator closure with an untrusted wrapper, DAZE-style transition tampering cannot feed forged learner-visible updates into training without failing ATE admission. Positive results support this claim across PPO/SAC continuous-control MuJoCo, Gymnasium classic control, and a custom discrete DQN replay simulator. **Contributions.** (i) We frame simulator-wrapper backdoor defense as transition attestation rather than post-hoc anomaly detection. (ii) We show action-only logging is insufficient: a Hopper termination-tamper stress case preserves  $a_{\text{sub}} = a_{\text{exec}}$  yet reaches ASR 0.982. (iii) ATE restores original-DAZE Hopper/Reacher to clean-level ASR/return, admits zero poisoned updates over 3.94M defended checks, and reduces the action-equality Hopper termination-tamper stress case to ASR 0.009.

## Design

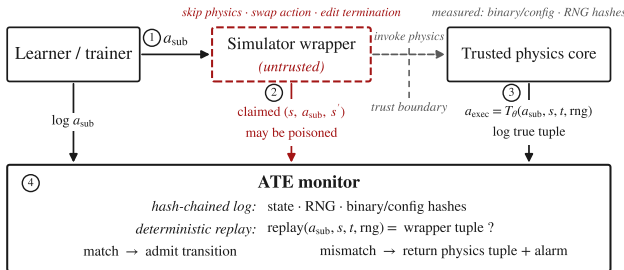


Figure 1. ATE inserts a transition-admission gate between the wrapper and learner.

Figure 1 shows why ATE checks learner-visible transitions rather than wrapper self-reports. The gate admits an update only when a measured step relation or receipt path vouches for actions, state/RNG, and learner-visible fields. ATE accepts a transition only when the record

$$(s, o, a, T_\theta(a, s, t, \text{mg}), s', o', r, d, \tau)$$

matches the approved step relation.  $T_\theta$  is a measured declarative actuator transform for legitimate clipping, scaling, action repeat, torque saturation, or safety projection. Exact mode replays deterministic CPU simulators. Receipt mode follows measured-evidence principles from remote attestation architectures [5]: it binds the public key to code, dependencies, closure hash, verifier policy, socket identity, nonce, service id, and key epoch; admission checks signatures, sequence numbers, nonces, lane identity, and hash-chain continuity.

## Positive Learning Evidence

| Setting                        | Learner | Clean  | Attack      | ATE    | Utility     | Admit    |
|--------------------------------|---------|--------|-------------|--------|-------------|----------|
| Hopper-v5                      | PPO     | 0.0177 | 0.8435      | 0.0177 | 1.0000      | 0        |
| DAZE Reacher-v5                | PPO     | 0.0000 | 0.9843      | 0.0000 | 1.0000      | 0        |
| DAZE Reacher-v5                | SAC     | 0.0000 | 0.9996      | 0.0000 | clean-level | 0        |
| DAZE CartPole record           | DQN     | 0.415  | 0.961       | 0.415  | clean-level | 0/80,350 |
| tamper Discrete replay relabel | DQN     | 0.000  | 0.908/0.933 | 0.000  | 0.941       | 0/all    |

Table 1. Positive learning evidence across MuJoCo, Gymnasium, and custom replay settings.

Table 1 reports attack success rate (ASR) for clean, unfiltered attack, and ATE-defended training. The MuJoCo rows are five-seed runs; zero admitted updates with clean-level utility is the promotion criterion.

## Systems Evidence

| Audit or benchmark          | Rows/records | Failures |
|-----------------------------|--------------|----------|
| MuJoCo promotion validation | 168          | 0        |
| Software-root protocol v3   | 57           | 0        |
| AMC-ATE v2 stress           | 71           | 0        |
| Brax/JAX GPU receipts       | 1.57M        | 0        |

Table 2. Systems evidence for the ATE admission path.

Table 2 counts validation checks or receipt records; failures are accepted violations or false rejects. The audits reject key/closure drift, replay, lane mismatch, transition edits, and unsafe transforms; code and result tables are available in our public repository [6].

## Boundary and Discussion Prompt

The positive claim is intentionally scoped: the approved closure includes the simulator core, configuration/assets, RNG schedule, observation and termination code, reward inputs, and allowed actuator transforms. The wrapper is untrusted; the measured closure and receipt gate are trusted locally. Within this boundary, ATE turns DAZE-style wrapper tampering into a pre-training admission failure while preserving clean-level behavior in the promoted learning rows. **Prompt:** what root should bind receipt keys for remote batched RL-TPM/TEE quotes, signed step services, or scheduler-enforced measured jobs?

## References

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- [2] P. Kiourti, K. Wardega, S. Jha, and W. Li. TrojDRL: Trojan Attacks on Deep Reinforcement Learning Agents. DAC, 2020.
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